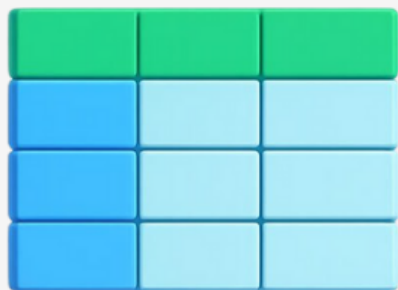


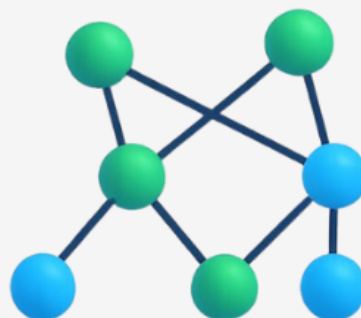
Relational DB vs Graph DB



Relational DB



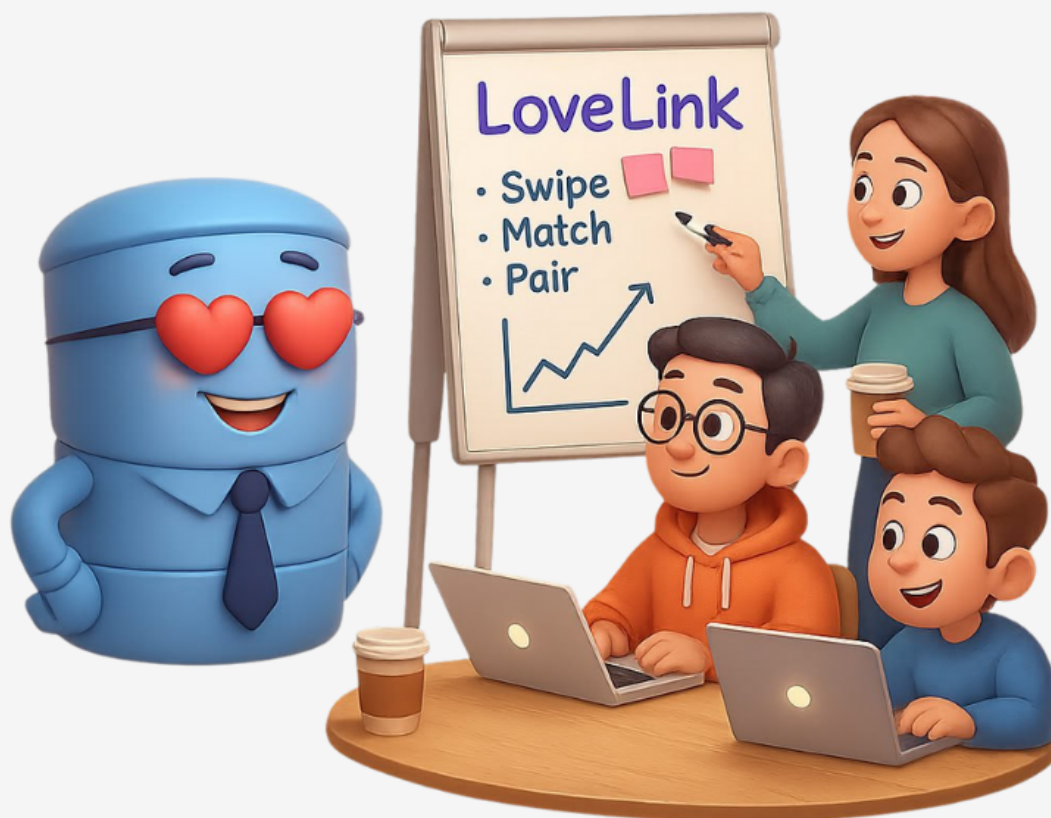
Graph DB



The Dating App Dilemma



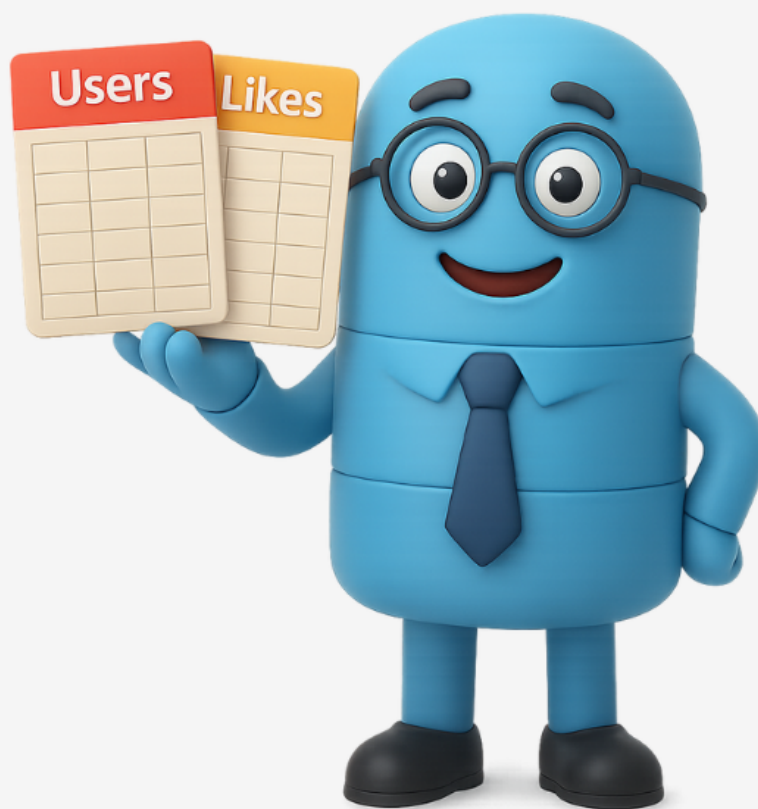
A **startup** wanted to build the **smartest dating app** ever. But they didn't know their biggest relationship problem... would be with their **database**. 😅



The Launch



They started with a **Relational Database** - good old **SQL**.
Perfect tables for **Users**, **Likes**, **Messages**
Fast, structured, reliable. 💾



The First Feature

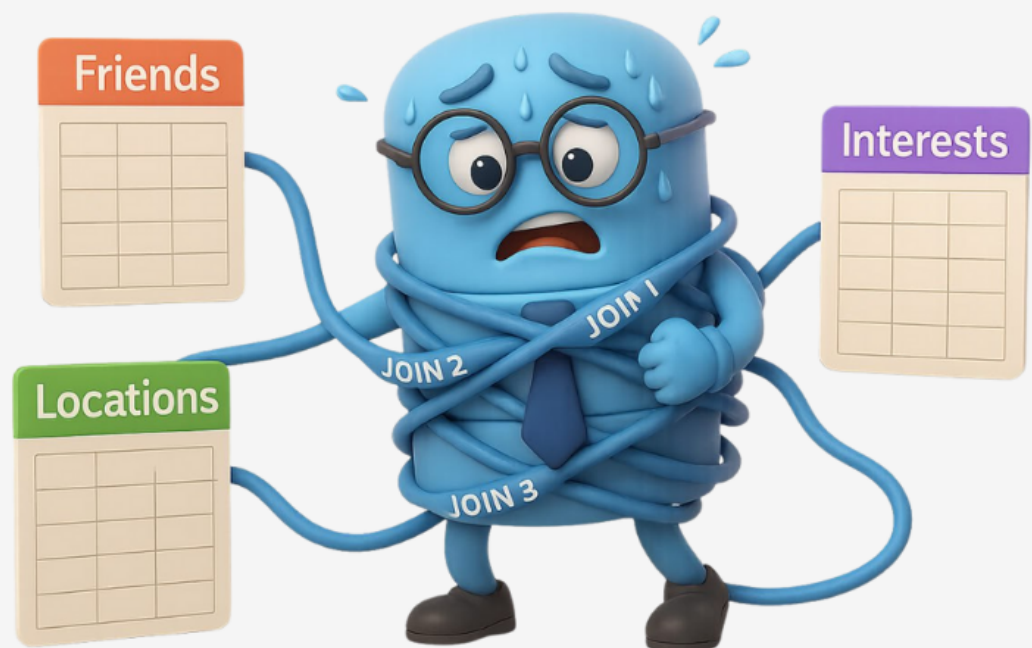


Then users asked:

“Show me friends of my friends who like hiking and live nearby.” 🏔️

The **SQL** team said: *“Sure... give us a moment.”*

(**4 JOINS** later...) The **server** is sweating. 🥵

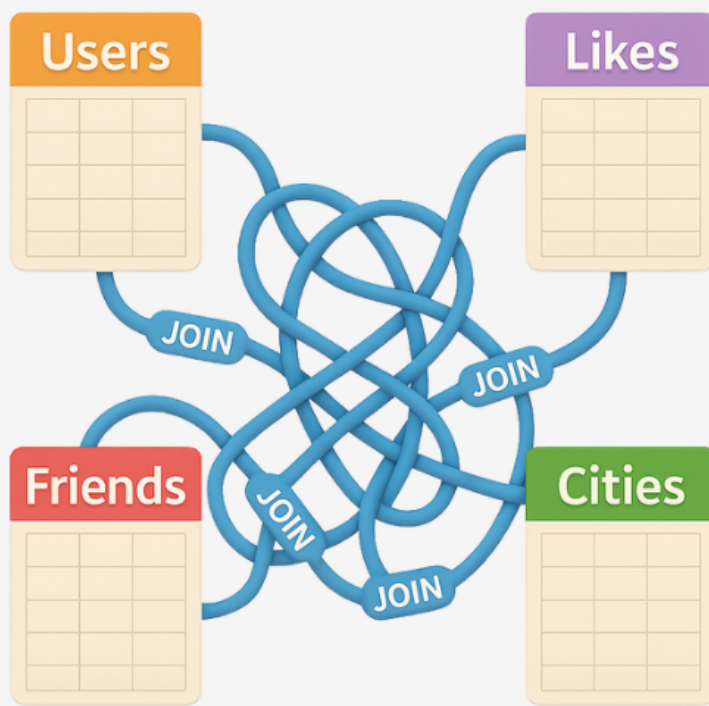


Why It Hurts?



In a **Relational DB**, connections live across many tables. To find relationships, you must **JOIN** them together – every time.

Great for **transactions** 👍, painful for **networks**. 😬



JOINS everywhere

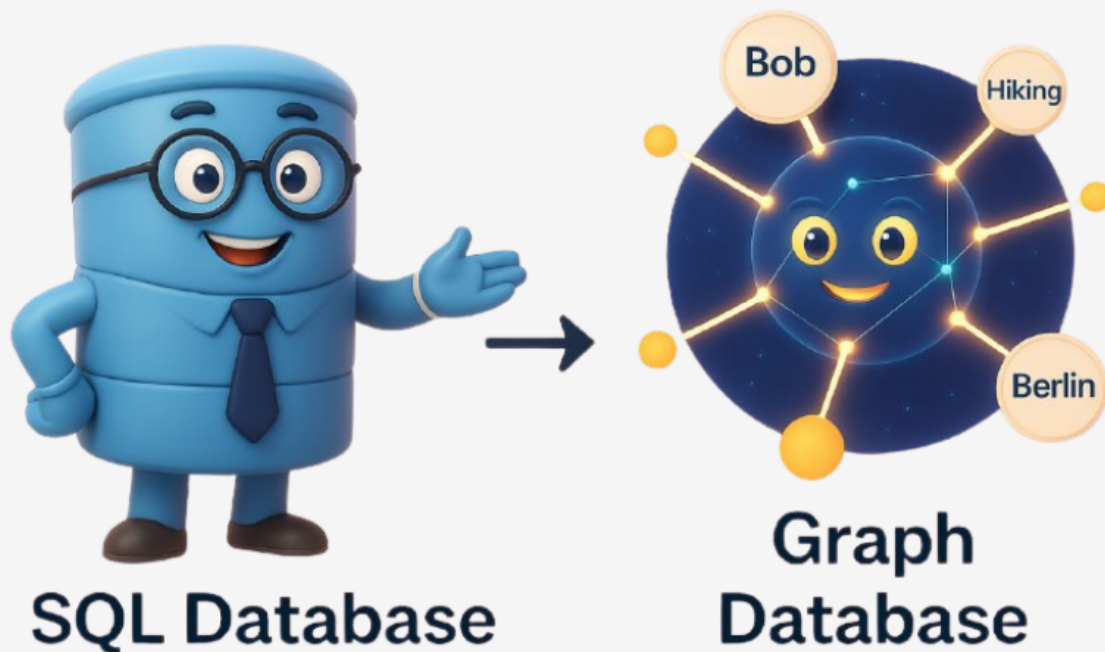


The Switch



So they switched to a **Graph Database**.
Suddenly, each **user** became a **node**, and **relationships** became **edges**.

Data didn't sit in **tables** anymore - it lived in a web.



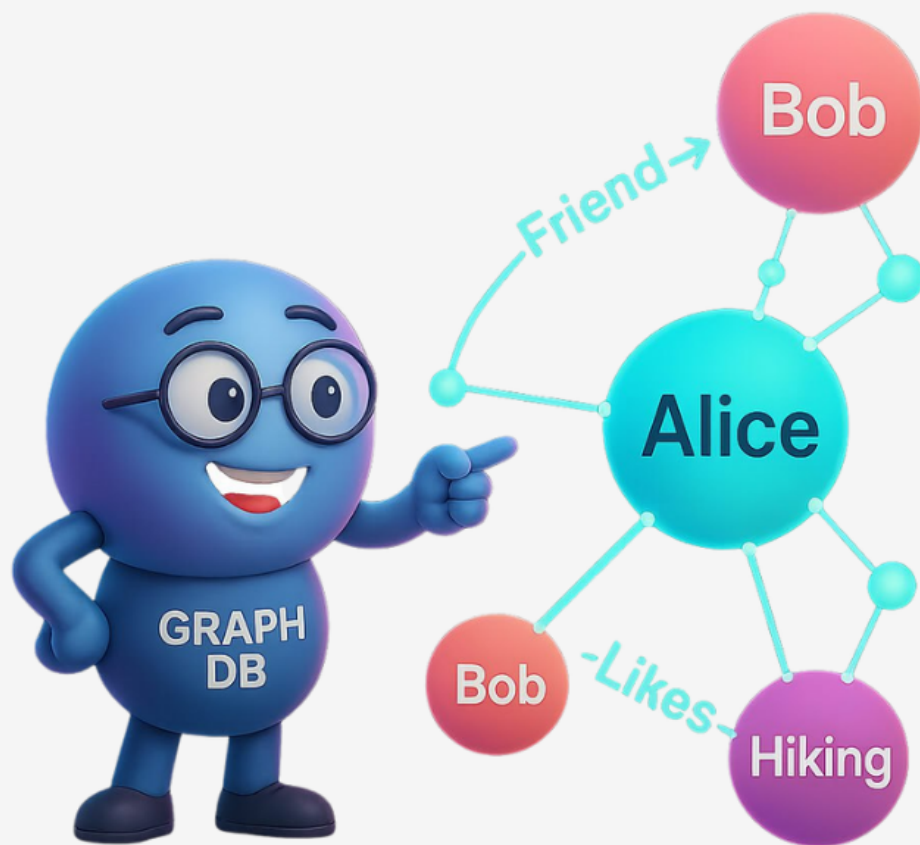
How It Thinks?



Graph DBs don't "**JOIN.**"

They traverse - simply follow the **connections.**

"Friends of Alice who like **Hiking**" = just walk the path



How do you talk to a Graph DB?

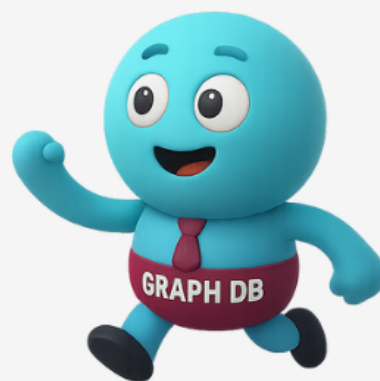


Unlike **SQL**, which speaks in **tables** and **JOINS**, **Graph DBs** use languages that speak in **connections**.

The most popular language is **Cypher** (used by Neo4j).

Instead of saying "SELECT users JOIN friends...", you write things like:

```
MATCH (Alice)-[:FRIEND]->(Bob)
```

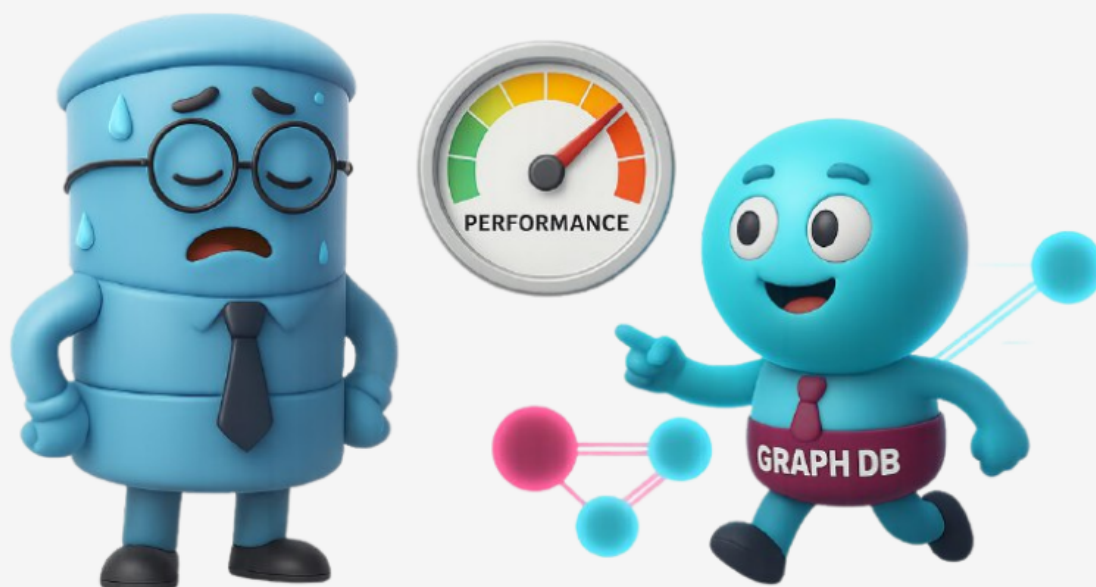


Performance



Result?

- Complex queries run **faster**
- Easier to model real-life **networks**
- Perfect for social **graphs**, fraud detection, recommendations



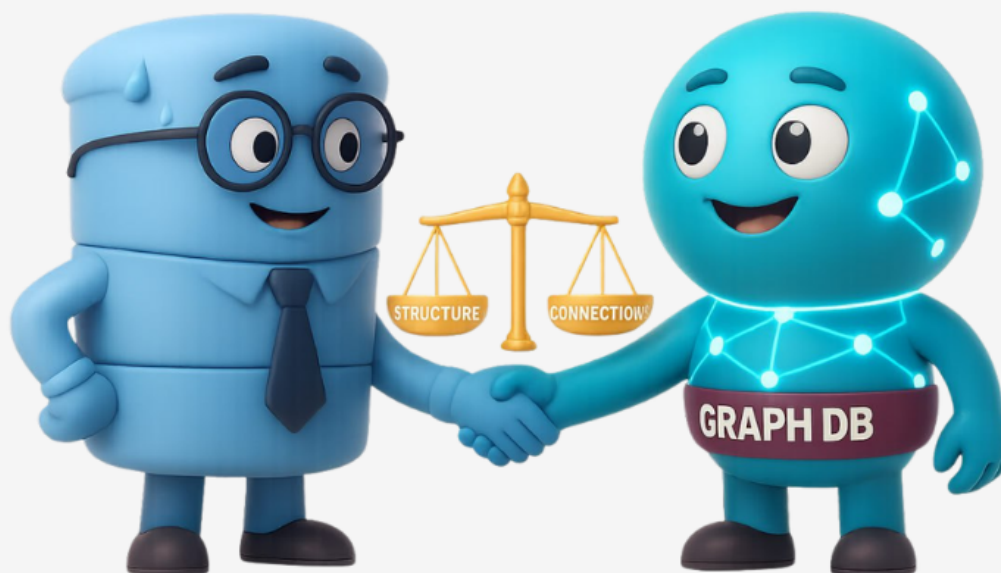
But Wait...



Relational DBs still rule for:

- ✓ Strong consistency
- ✓ Complex transactions
- ✓ Traditional business data

Graph DBs shine when relationships matter most.

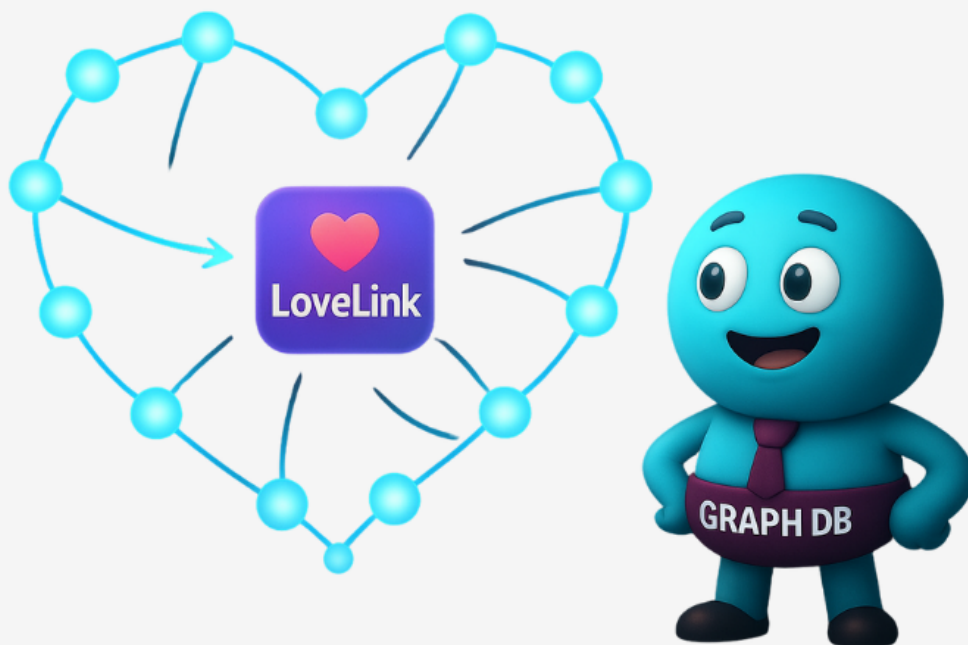


The Happy Ending



The **app** finally worked like love itself -
powered by **connections**. ❤️

Graph DBs don't just store **data**...
they understand how things are related. 🕸️



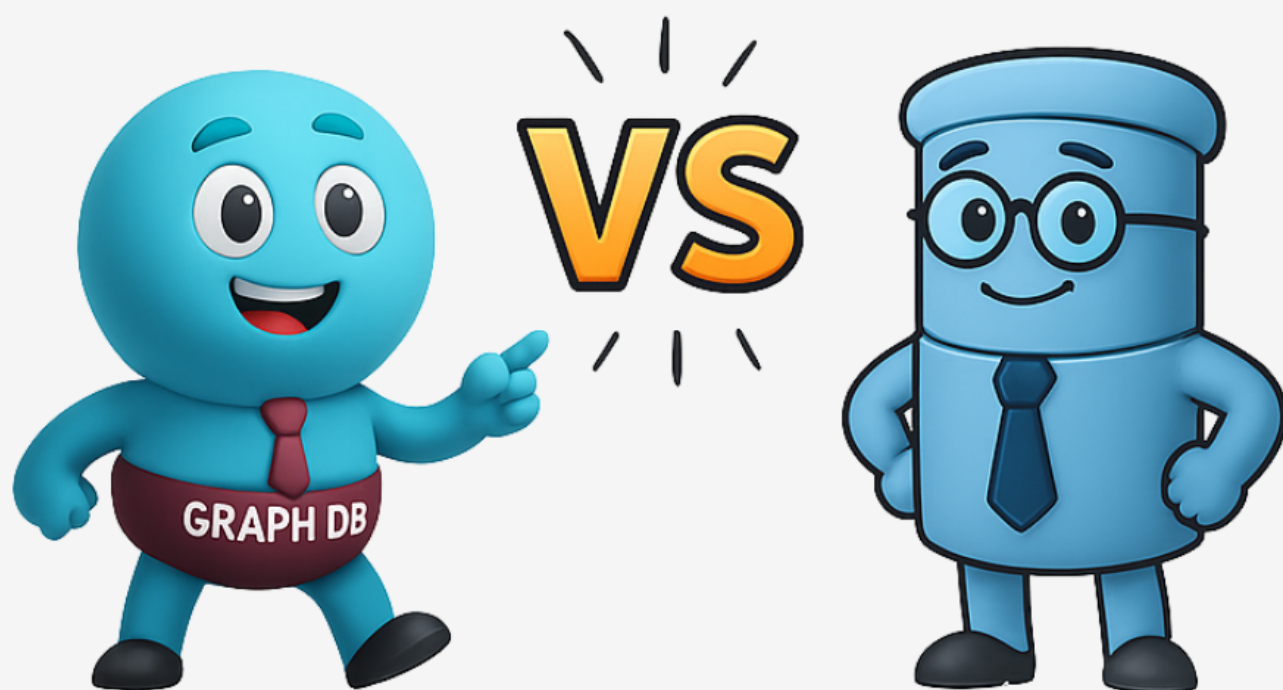
Summary



Aspect	Relational D	Graph DB
Structure	Tables	Nodes & Edges
Query	SQL	Cypher / Gremlin
Strength	Transactions, tabular data	Relationships, traversal
Weakness	Complex joins	Aggregations
Schema	Rigid	Flexible



Graph DB or Relational DB Which side are you on?



➡ Drop your choice & why
in the comments 💬👉

